

Amphibious Combat Vehicle (ACV) Crew Gunnery Trainer (CGT)



29 April 2025

Amphibious Combat Vehicle (ACV) Crew Gunnery Trainer (CGT)

Request for White Papers

1. **Project Number:** M67854-26-9-0004
2. This Request for White Papers (RWP) is being issued to seek participation in a prototype project pursuant to the Other Transaction Authority (OTA) authority granted under 10 U.S.C. § 4022, which enables the Department of Defense (DoD) to carry out certain prototype projects and award follow-on production agreements. OTAs were created to give DoD the flexibility necessary to adopt and incorporate business practices that reflect commercial industry standards and best practices into its transactions and agreements.
3. In addition to issuing agreements for prototype projects, the United States Marine Corps (USMC), Program Executive Officer Land Systems (PEO LS), Program Manager Advanced Amphibious Assault (PM AAA) anticipates leveraging the follow-on production award authority in 10 U.S.C. § 4022(f) for the successful prototypes. The full training solutions being considered for follow-on production include all the training material and equipment, personnel, and logistics support each of (2) training systems in support of ACV gunnery. Government estimates for the ACV CGT prototype and development phase is estimated to be in the \$5M range with an option for an additional estimated, \$7M for software development. The anticipated follow-on production contract is currently estimated in the \$12M range. The availability of resources and funding, as well as the quality of solution papers, will affect the number of agreements PM AAA will award. The Government may ultimately award one, multiple, or no agreements whatsoever in support of the ACV CGT in its sole discretion.
4. The ACV platform is currently being produced by BAE Systems (BAE) under competitively awarded contract M67854-16-C-0006. The Government does *not* currently own the required rights to the ACV-Personnel (ACV-P) and ACV 30-millimeter cannon (ACV-30) Technical Data Package (TDP). This will require participants in the prototype project to scan and 3-D model the required components and their constituent parts if that vendor's White Paper Response results in an award. The Government will provide access to an ACV-P and ACV-30 (vehicles) as Government Furnished Equipment (GFE) to the successful Contractor for the purpose of conducted the scanning and modeling efforts for any prototype project participant. PM AAA desires that the prototype project and systems maintain commonality with the ACV Suite of Training Systems' use of Unreal Gaming Engine 5 software for standardization across all training systems. The Government intends to make software already developed via a previous effort built upon Unreal Gaming Engine 5 available to the prototype participants upon the start of the project that will accelerate development of the CGT prototype.
 - a. This software creates a virtual reality environment for the user that consists of surf zone and land driving scenarios and their accompanying physics models. The software also includes 3-D modeling of the outside of the ACV-P variant and modeling of the driver's station to include switch functionality. The existing Driver Training System incorporates a realistic Unreal Gaming Engine environment that replicates the surf zone environment for students as part of its operating baseline. The Program Manager

is seeking standardization across all training platforms to gain increased training value in terms of safety, training efficiency and effectiveness, and platform familiarization. Use of Unreal Gaming Engine software for this project is highly encouraged by the Government due to future sustainment plans for the entirety of the ACV Suite of Training Systems.

5. Points of Contact:

Contact Information	Project Officer	Agreements Officer	Training Analyst
Name:	Major Alex Puraty	Shanna Laging	Capt Mike Fanning
Email Address:	alexander.puraty@usmc.mil	shanna.laging@usmc.mil	Michael.b.fanning.mil@usmc.mil

6. Problem Statement:

PM AAA has a requirement for the development and acquisition of training systems for ACV gunnery. The legacy Assault Amphibian Vehicle (AAV) had an assigned training system for crews to practice gunnery in a simulated environment. Currently, the ACV-P and ACV 30 variants have no training systems for crews to practice gunnery at any level.

The purpose of this prototype project will be for the participants to develop a software-based training system with Commercial Off-The-Shelf (COTS) hardware for a mixed reality simulation-based training solution focused on developing vehicle crews and their gunnery skills in a realistic, immersive virtual reality environment.

Production: The Government anticipates a production procurement of 24-36 systems; with 2-4 deployable variants of the CGT, but this may change depending on Government funding availability. Development of software is the priority.

7. Desired Objective:

The prototype participants shall develop and deliver one prototype CGT based on their efforts during the competitive prototyping phase. The CGT will use a mixed reality system to allow ACV-P crews to conduct the Gate to Live-Fire Table II which consists of simulated Tables IV, V, and VI per NAVMC 3500.2D, Assault Amphibian Training and Readiness Manual.

Prototyping phase will include the development of a virtual immersive environment of Table II with simulated fire and maneuver from both friendly and threat units. Software development is the priority of effort. The user experience should include communications system for vehicle intercom, Day/Night capability with thermal ID of friendly and threat units. The learning management system should be able to record time to target, time to fire, and time to destroy threat. Learning management system/Artificial Intelligence should be capable of controlling threat unit actions, simulate adjacent friendly units, and simulate higher headquarters instructions. Learning management system should be capable of facilitating instructor after action reviews in the form of video and audio communications playback of engagements. The Government additionally wants a function that allows solo gunner

training within the system to develop individual required skills to participate with the crew. The CGT COTS hardware should incorporate a motion base for the crew.

The plan for the CGT is to begin formal design and testing activities in First Quarter, FY 2026 (1QFY26) with completion of competitive prototyping approximately Fourth Quarter, FY 2026 (4QFY26).

The prototype software and hardware baseline should also accommodate and consider the following future tasks:

- Optional development includes the development of Tables VII-XII. Further development of the communications system will include vehicle to vehicle, simulated higher headquarters, simulated ground maneuver forces.
- The environment will include simulated friendly units maneuver, fires, and communications.
- It should allow the development of threat vehicles and threat tactics for free play scenarios as well as force-on-force capability development with associated closed network capability.
- It should also allow for the addition of the ACV-30 virtual and hardware.

8. Government's Intent with Respect to Rights in the Responding Party's Technical Data and Computer Software:

The Government intends to acquire rights in the Responding Party's technical data and computer software (data rights) developed under this prototype effort. The Government will also consider the rights/terms associated with constituent third-party solutions in making its determination. In order to support the anticipated development of the ACV CGT (including all components and computer software), the Government anticipates that it will require the delivery of the Responding Party's tech data and computer software with unlimited or Government purpose rights to transfer and use such delivered data in future developments, which may be performed by the Government or a third party.

The Government intends to take the flexibility offered by the vendor's preferred data rights into account when analyzing proposals. The Government is open to negotiating the specific terms; however, any Responding Party unwilling to offer the Government sufficient data rights to meet the Government's developmental needs should consider whether it wishes to proceed in responding to the Government's request. Therefore, a vendor's response to this RWP must confirm its commitment to delivering any prototype software or device developed under this effort with Unlimited or Government purpose rights. Additionally, the vendor should describe any restrictions accompanying any third-party software and hardware solutions that it proposes as part of its Response. Further, the Responding Party must conspicuously call out, in its response, any restrictions in the transfer and use of data that affect or impact the Government's stated intent as well as any unknowns presented by the incorporation of third-party hardware and software.

9. Process:

Phase I Request for White Papers

White papers, exclusive of attachments, should be no longer than 10 pages and must be submitted in a searchable PDF using no smaller than Times New Roman 12 pt. font. Text should be double-

spaced, and the document should contain section headers and page numbers for ease of reference. Margins must be printable (generally no less than ½ inch) on each edge. The Responding Party may add a cover letter and/or acronym list which will not count against the page count. The Responding Party is discouraged from adding additional attachments other than those requested but may use their discretion to attach a reasonable additional number of pages to supplement their white paper as an attachment. The USMC is under no obligation to read any such additional attachments and those who include additional information in an attachment are encouraged to clearly explain the significance of the attachment within the standard page count.

Vendor white papers are due via email to the Contract Specialists, Ms. Hana Musser at hana.p.musser.civ@usmc.mil and Mr. Matthew Mercer at matthew.mercer@usmc.mil, and the Agreements Officer Ms. Shanna Laging at shanna.laging@usmc.mil **no later than 0900 EST on Thursday, 29 May 2025**. Due to email memory size limitations, the USMC can only receive emails less than 10MB. If your submission is over 10MB, please submit electronically to Ms. Shanna Laging at shanna.laging@usmc.mil via the Safe Access File Exchange (SAFE) website provided at <https://safe.apps.mil/>. Title the subject line of the email as follows:

“Company_Name_ACV CGT_Response_Request for White Papers”

The USMC requests that interested vendors provide the following information:

1. A description of how the vendor will approach developing the ACV CGT in accordance with Enclosure 1.
2. A description of the vendor’s software (SW) development approach and a list and description of any use restrictions associated with vendor proprietary or third -party technical data or software being proposed to accomplish the effort. The vendor should also mention if it needs software or technical data from the Government.
3. Rough Order Magnitude (ROM) pricing and a schedule for design and delivery of one CGT prototype system. Vendors should separately price optional development efforts as described in Paragraph 7 in the Section above. Provide ROM in an Excel spreadsheet (that can be edited) and which is not considered part of the page count but should be marked and separately numbered as Attachment C-1.
4. Description of the scalability of the vendor’s design to incorporate the ACV-P, ACV-30, as well as a deployable CGT variant for future incorporation.
5. Description of the vendor’s anticipated approach and production capability over a period of 6-12 months to facilitate the delivery of the prototype system and for any potential follow-on production efforts.
6. Description of the vendor’s Learning Management System, to include reports and ability to offload/transfer records of students by crew and by individual training.
7. If the vendor’s system is currently offered, copies of the applicable terms and conditions of sale, to include *all* licensing agreements. Terms and conditions and licensing agreements are not considered part of the page count and should be marked and separately labeled as Attachment B. If offerors have more than one document, a table of contents for Attachment B is recommended.
8. Description of the basis for the vendor’s status as a small business or non-traditional defense contractor, if applicable. If the vendor does not qualify as a small business, please

provide a response that details how at least one “non-traditional defense contractor” is participating to a significant extent. Any response provided should include a qualitative and quantitative narrative about the contributions of the “non-traditional defense contractor,” as well as a response that describes how the identified “non-traditional defense contractor” who will participate in a “significant extent” qualifies as a “non-traditional defense contractor.” For this effort, “significant extent” is defined as one of the following:

- The participation causes a material reduction in the cost or schedule; or
 - The participation causes an increase in the performance of prototype; or
 - The performer is responsible for a new key component, technology, or process on the critical path
9. Please identify any sub-contractors that will be assisting in any capacity with the project, use Attachment C to quantify their efforts. Sufficient background on previous projects should be provided to enable the Government to fairly justify their inclusion in the project.
 10. Please identify any Foreign Partners that you anticipate being involved that may require official approval from their country’s government. If a company is owned wholly or partially by foreign persons/entities, they will be required to complete a Certificate Pertaining to Foreign Interests (i.e., SF 328). Additional information can be found in the National Industrial Security Program Operating Manual (NISPOM) Rule – <https://www.dcsa.mil/mc/isd/NISPOM-Rule/>
 11. If applicable, please identify the facility clearance process and if there are any International Traffic in Arms Regulations (ITAR) restrictions in effect.

Phase II: Selection and Award

After the USMC reviews the white papers, it intends to evaluate the responses to determine which responses provide the best approach in terms of schedule, cost, and performance risk and most likely will succeed in producing the best solution to the prototype problem. As a procedural matter, the Government will likely exclude vendors based on their responses only without further communication. However, the Government may identify those companies with approaches from which it wishes to seek more information and communicate with those companies only. The Government may seek more detailed written documentation relating to those specific company’s (-ies’) approach regarding the cost, performance, or schedule associated with the testing and demonstration of prototypes or production. The Government also anticipates leveraging the flexibility that OT authority provides to collaboratively engaging with companies in the development of the statement of work. This flexibility includes modifying the anticipated path forward.

Questions or comments regarding this Request for White Papers may be addressed via email, to the Contract Specialists, Ms. Hana Musser at hana.p.musser.civ@usmc.mil and Mr. Matthew Mercer at matthew.mercer@usmc.mil, and the Agreements Officer, Ms. Shanna Laging at shanna.laging@usmc.mil. Submit all questions and comments no later than **0900 EST on Wednesday, 14 May 2025**. Title the subject line of the email as follows:

“Company Name_ACV CGT_Questions_Request for White Papers”

10. Terms and Conditions

1. The Government will not pay any costs associated with the preparation of white papers, subsequent submissions, or negotiations. Any Government obligation to make a payment will only occur as the result of a written agreement between the Government and a party, which the Government will pursue at its sole discretion.
2. The Government, to the extent permitted by law and regulation, will safeguard and treat information obtained pursuant to the Contractor's disclosure as confidential where the information has been marked "confidential" or "proprietary" by the company. To the extent permitted by law and regulation, such information will not be released by the Government to the public pursuant to a Freedom of Information Act request, 5 U.S.C. Section 552, without prior notification to the Contractor. The Government may transfer documents provided by the Contractor to any department or agency within the Executive Branch if the information relates to matters within the organization's jurisdiction.
3. All respondents shall be registered in the System for Award Management (www.sam.gov).
4. FAR 52.204-24 Representation Regarding Certain Telecommunications and Video Surveillance Services or Equipment (Nov 2021)

The Offeror shall not complete the representation at paragraph (d)(1) of this provision if the Offeror has represented that it "does not provide covered telecommunications equipment or services as a part of its offered products or services to the Government in the performance of any contract, subcontract, or other contractual instrument" in paragraph (c)(1) in the provision at 52.204-26, Covered Telecommunications Equipment or Services—Representation, or in paragraph (v)(2)(i) of the provision at 52.212-3, Offeror Representations and Certifications-Commercial Products or Commercial Services. The Offeror shall not complete the representation in paragraph (d)(2) of this provision if the Offeror has represented that it "does not use covered telecommunications equipment or services, or any equipment, system, or service that uses covered telecommunications equipment or services" in paragraph (c)(2) of the provision at 52.204-26, or in paragraph (v)(2)(ii) of the provision at 52.212-3.

(a) Definitions. As used in this provision—

Backhaul, covered telecommunications equipment or services, critical technology, interconnection arrangements, reasonable inquiry, roaming, and substantial or essential component have the meanings provided in the clause 52.204-25, Prohibition on Contracting for Certain Telecommunications and Video Surveillance Services or Equipment.

(b) Prohibition.

(1) Section 889(a)(1)(A) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (Pub. L. 115-232) prohibits the head of an executive agency on or after August 13, 2019, from procuring or obtaining, or extending or renewing a contract to procure or obtain, any equipment, system, or service that uses covered telecommunications equipment or services as a

substantial or essential component of any system, or as critical technology as part of any system. Nothing in the prohibition shall be construed to—

(i) Prohibit the head of an executive agency from procuring with an entity to provide a service that connects to the facilities of a third-party, such as backhaul, roaming, or interconnection arrangements; or

(ii) Cover telecommunications equipment that cannot route or redirect user data traffic or cannot permit visibility into any user data or packets that such equipment transmits or otherwise handles.

(2) Section 889(a)(1)(B) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (Pub. L. 115-232) prohibits the head of an executive agency on or after August 13, 2020, from entering into a contract or extending or renewing a contract with an entity that uses any equipment, system, or service that uses covered telecommunications equipment or services as a substantial or essential component of any system, or as critical technology as part of any system. This prohibition applies to the use of covered telecommunications equipment or services, regardless of whether that use is in performance of work under a Federal contract. Nothing in the prohibition shall be construed to—

(i) Prohibit the head of an executive agency from procuring with an entity to provide a service that connects to the facilities of a third-party, such as backhaul, roaming, or interconnection arrangements; or

(ii) Cover telecommunications equipment that cannot route or redirect user data traffic or cannot permit visibility into any user data or packets that such equipment transmits or otherwise handles.

(c) Procedures. The Offeror shall review the list of excluded parties in the System for Award Management (SAM) (<https://www.sam.gov>) for entities excluded from receiving federal awards for "covered telecommunications equipment or services".

(d) Representation. The Offeror represents that—

(1) It ☐ will, ☐ will not provide covered telecommunications equipment or services to the Government in the performance of any contract, subcontract or other contractual instrument resulting from this solicitation. The Offeror shall provide the additional disclosure information required at paragraph (e)(1) of this section if the Offeror responds "will" in paragraph (d)(1) of this section; and

(2) After conducting a reasonable inquiry, for purposes of this representation, the Offeror represents that—

It ☐ does, ☐ does not use covered telecommunications equipment or services, or use any equipment, system, or service that uses covered telecommunications equipment or services. The Offeror shall provide the additional disclosure information required at paragraph (e)(2) of this section if the Offeror responds "does" in paragraph (d)(2) of this section.

(e) Disclosures.

(1) Disclosure for the representation in paragraph (d)(1) of this provision. If the Offeror has responded "will" in the representation in paragraph (d)(1) of this provision, the Offeror shall provide the following information as part of the offer:

(i) For covered equipment—

(A) The entity that produced the covered telecommunications equipment (include entity name, unique entity identifier, CAGE code, and whether the entity was the original equipment manufacturer (OEM) or a distributor, if known);

(B) A description of all covered telecommunications equipment offered (include brand; model number, such as OEM number, manufacturer part number, or wholesaler number; and item description, as applicable); and

(C) Explanation of the proposed use of covered telecommunications equipment and any factors relevant to determining if such use would be permissible under the prohibition in paragraph (b)(1) of this provision.

(ii) For covered services—

(A) If the service is related to item maintenance: A description of all covered telecommunications services offered (include on the item being maintained: Brand; model number, such as OEM number, manufacturer part number, or wholesaler number; and item description, as applicable); or

(B) If not associated with maintenance, the Product Service Code (PSC) of the service being provided; and explanation of the proposed use of covered telecommunications services and any factors relevant to determining if such use would be permissible under the prohibition in paragraph (b)(1) of this provision.

(2) Disclosure for the representation in paragraph (d)(2) of this provision. If the Offeror has responded "does" in the representation in paragraph (d)(2) of this provision, the Offeror shall provide the following information as part of the offer:

(i) For covered equipment—

(A) The entity that produced the covered telecommunications equipment (include entity name, unique entity identifier, CAGE code, and whether the entity was the OEM or a distributor, if known);

(B) A description of all covered telecommunications equipment offered (include brand; model number, such as OEM number, manufacturer part number, or wholesaler number; and item description, as applicable); and

(C) Explanation of the proposed use of covered telecommunications equipment and any factors relevant to determining if such use would be permissible under the prohibition in paragraph (b)(2) of this provision.

(ii) For covered services—

(A) If the service is related to item maintenance: A description of all covered telecommunications services offered (include on the item being maintained: Brand; model number, such as OEM number, manufacturer part number, or wholesaler number; and item description, as applicable); or

(B) If not associated with maintenance, the PSC of the service being provided; and explanation of the proposed use of covered telecommunications services and any factors relevant to determining if such use would be permissible under the prohibition in paragraph (b)(2) of this provision.

(End of provision)

5. FAR 52.204-25 will be incorporated by reference into the final agreement.

Enclosure 1: Top 10 Crew Gunnery Trainer Specifications

1. The CGT stations shall model the ACV driver, gunner, and vehicle commander stations.
2. The CGT shall model both the ACV-P weapon station and ACV-30 for gunnery.
3. The CGT shall virtually model all AA Gunnery Tables per NAVMC 3500.2D.
4. The CGT shall be able to simulate friendly and enemy fires.
5. The CGT shall be capable of communications at the crew, section, platoon, and company level.
6. The CGT shall be capable of simulating force-on-force training between all systems.
7. The CGT shall have an after-action review function.
8. The CGT IOS shall allow the instructor to create scenarios of varying difficulty.
9. The CGT IOS shall allow the instructor to export communications and marksmanship data from each of the scenarios ran in the CGT.
10. The CGT shall be a transportable system.